Math 342W / 642 / RM742 Spring 2026
Midterm Examination Two

Professor Adam Kapelner

May 13, 2026

Full Name ______________________________

Code of Academic Integrity

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Cheating Using or attempting to use unauthorized assistance, material, or study aids in examinations or other academic work or preventing, or attempting to prevent, another from using authorized assistance, material, or study aids. Example: using an unauthorized cheat sheet in a quiz or exam, altering a graded exam and resubmitting it for a better grade, etc.

I acknowledge and agree to uphold this Code of Academic Integrity.

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signature date

Instructions

This exam is 110 minutes (variable time per question) and closed-book. You are allowed **two** pages (front and back) of a "cheat sheet", blank scrap paper (provided by the proctor) and a graphing calculator (which is not your smartphone). Please read the questions carefully. Within each problem, I recommend considering the questions that are easy first and then circling back to evaluate the harder ones. No food is allowed, only drinks.

Problem 1 You are a credit card company looking to acquire new customers through cold mailings: you send a "junk mail" offering to people who have no prior relationship with. The price of the mailing is \$1. The company has determined that the net lifetime value (LTV) of a new customer is \$500 on average. Let $y_i = 1$ denote that person i has signed up to be a new credit card customer from the mailing. Let $\hat{y}_i = 1$ denote that your model predicts that person i will sign up to be a new credit card customer from the mailing

(a) [1 pt / 1 pts] What is the cost of a false positive (c_{FP})? Include the unit.

\$1

(b) [1 pt / 2 pts] What is the cost of a false negative (c_{FN})? Include the unit.

\$500

(c) [3 pt / 5 pts] If you were working at this credit card company, would you create a asymmetric cost prediction model? Why or why not?

Yes. Since $c_{FP} \neq c_{FN}$, you should use an asymmetric cost objective when fitting the model.

(d) [4 pt / 9 pts] As a function of $y_1, \ldots, y_n, \hat{y}_1, \ldots, \hat{y}_n, n$ and the answers to the two previous questions, write an expression that computes the in sample AC (average cost per prediction).

$$AC = \frac{1}{n} \sum_{i=1}^{n} \$500 \mathbb{1}_{y_i=1, \hat{y}_i=0} + \$1 \mathbb{1}_{y_i=0, \hat{y}_i=1}$$

Regardless of what you wrote in the previous questions, you employ a probit regression. The results are found on the next page.

(e) [4 pt / 13 pts] Approximate the probability of a future person opening a credit card. This new person's characteristics are given by $\boldsymbol{x}_\star = \boldsymbol{0}_p^\top$.

$$\begin{aligned}
\hat{y}_\star &= \boldsymbol{b}\boldsymbol{x}_\star = \boldsymbol{b}[1 \ \boldsymbol{0}_p^\top] = b_0 = -3.994 \\
\hat{p}_\star &= \Phi(\hat{y}_\star) = \Phi(-3.994) \approx 0 \quad \text{where } \Phi \text{ is the CDF for } Z \sim \mathcal{N}(0,\, 1)
\end{aligned}$$

(Intercept)	age
-3.994e+00	1.476e-02
workclassLocal-gov	workclassPrivate
-3.959e-01	-2.881e-01
workclassSelf-emp-inc	workclassSelf-emp-not-inc
-1.807e-01	-5.582e-01
workclassState-gov	workclassWithout-pay
-4.591e-01	-4.192e+00
fnlwgt	education11th
3.544e-07	7.451e-02
education12th	education1st-4th
2.374e-01	-3.662e-01
education5th-6th	education7th-8th
-2.723e-01	-2.933e-01
education9th	educationAssoc-acdm
-1.284e-01	7.181e-01
educationAssoc-voc	educationBachelors
7.167e-01	1.078e+00
educationDoctorate	educationHS-grad
1.618e+00	4.267e-01
educationMasters	educationPreschool
1.268e+00	-1.022e+01
educationProf-school	educationSome-college
1.588e+00	6.294e-01
marital_statusMarried-AF-spouse	marital_statusMarried-civ-spouse
1.471e+00	1.057e+00
marital_statusMarried-spouse-absent	marital_statusNever-married
1.875e-02	-2.548e-01
marital_statusSeparated	marital_statusWidowed
-6.475e-02	1.004e-01
occupationArmed-Forces	occupationCraft-repair
-6.783e-01	2.770e-02
occupationExec-managerial	occupationFarming-fishing
4.589e-01	-5.835e-01
occupationHandlers-cleaners	occupationMachine-op-inspct
-3.875e-01	-1.703e-01
occupationOther-service	occupationPriv-house-serv
-4.302e-01	-2.128e+00
occupationProf-specialty	occupationProtective-serv
3.036e-01	3.475e-01
occupationSales	occupationTech-support
1.684e-01	3.686e-01
occupationTransport-moving	relationshipNot-in-family
-5.239e-02	1.150e-01

(f) [1 pt / 14 pts] Given $\mathbb{D}$ and the output from the probit regression on the previous page but no further information, would you be able to trace out an ROC curve? Yes / no

(g) [1 pt / 15 pts] Given $\mathbb{D}$ and the output from the probit regression on the previous page but no further information, would you be able to trace out a DET curve? Yes / no

(h) [1 pt / 16 pts] Given $\mathbb{D}$ and the output from the probit regression on the previous page but no further information, would you be able to construct the confusion matrix of interest? Yes / no (This question was ambiguous)

(i) [1 pt / 17 pts] Given $\mathbb{D}$ and the output from the probit regression on the previous page but no further information, would you be able to trace out an ROC curve for the same model but now fit with a logistic link instead of a probit link? Yes / no

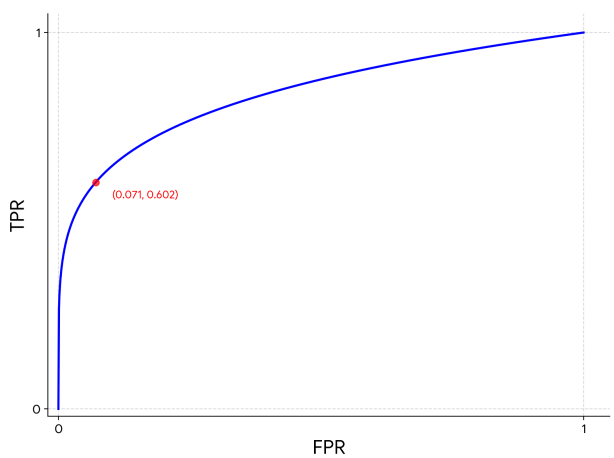
(j) [1 pt / 18 pts] Given $\mathbb{D}$ and the output from the probit regression on the previous page but no further information, would you be able to trace out a DET curve for the same model but now fit with a logistic link instead of a probit link? Yes / no

Regardless of what you wrote in the previous yes/no questions, consider the following confusion matrix:

```
   y_hat
y       0     1
  0 21056  1598
  1  2988  4520
```

(k) [5 pt / 23 pts] Draw the ROC curve *as best you can* given this information. Label your axes. Label critical points on the axes. Hint: there are many correct answers.

From the confusion matrix, we only can draw one point: $< FPR = 1598/(21056 + 1598) = 0.071,\ TPR = 4520/(2988 + 4520) = 0.602 >$. However, since we know the ROC must be a monotonically increasing function starting at $< 0, 0 >$ and ending at $< 1, 1 >$ passing through this point, we can estimate the curve a la what is below:



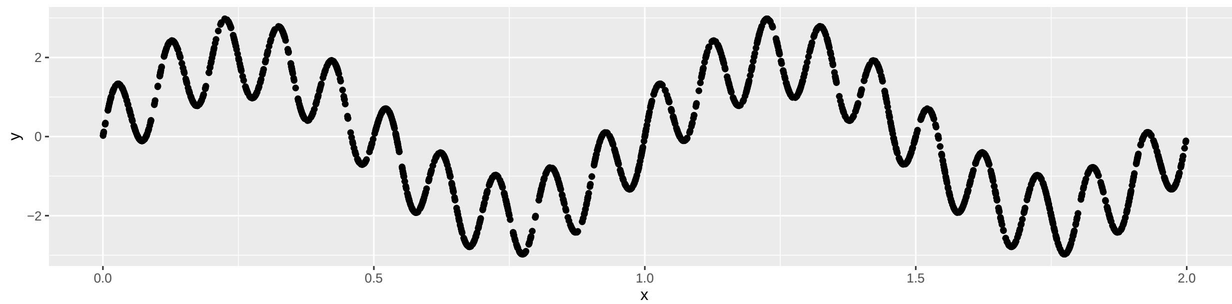
(l) [2 pt / 25 pts] Estimate the AUC from the previous question.

This will depend on how you drew the curve, but it is difficult to justify an answer outside of the range $[0.6, 0.9]$.

(m) [2 pt / 27 pts] If the cost of a false negative is very large, what can you do to improve this classifier?

Lower the threshold p_{th}

Problem 2 Consider the following $\mathbb{D}$ with $n = 1,000,\ p = 1$.



(a) [9 pt / 36 pts] Which one(s) of the following statement(s) are most likely true?

i) $f(x) = 0$

ii) a linear model will fit $f(x)$ well

iii) a linear model with polynomial terms of degree 2 will fit $f(x)$ well

iv) a linear model with polynomial terms of degree 2 can *only* be fit if one uses three splits of $\mathbb{D}$ — one for training, one for selection and one for testing

v) TRUE a linear model with polynomial terms of degree d where d is selected can *only* be fit and validated honestly if one uses three splits of $\mathbb{D}$ — one for training, one for selection and one for testing

vi) TRUE a linear model with polynomial terms will take many degrees of freedom to fit $f(x)$ well

vii) TRUE a linear model with polynomial terms with degrees of freedom that fit $f(x)$ well will suffer from Runge's phenomenon

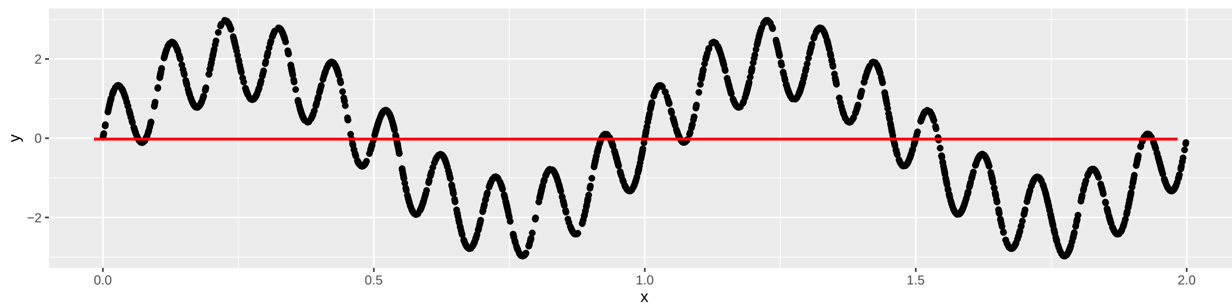
viii) TRUE a linear model with polynomial terms with degrees of freedom that fit $f(x)$ well will outperform a tree model during future interpolations

ix) a linear model with polynomial terms with degrees of freedom that fit $f(x)$ well will predict well for $x > 2$

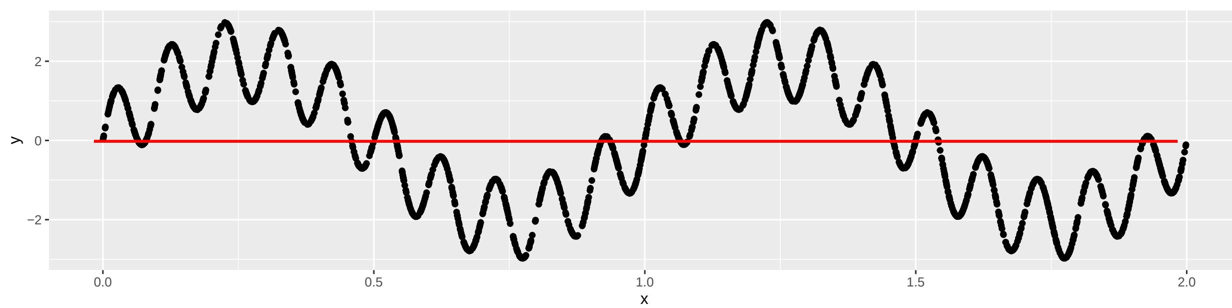
(b) [4 pt / 40 pts] Provide an $\mathcal{H}$ that will fit this model optimally. Use set-builder notation appropriately. Include an intercept term w_0. Label the parameters $w_0, w_1, w_2, \ldots$

$$\mathcal{H} = \left\{ w_0 + w_1 \sin\left(w_2 + w_3 x\right) + w_4 \sin\left(w_5 + w_6 x\right) \;:\; \boldsymbol{w} \in \mathbb{R}^7 \right\}$$

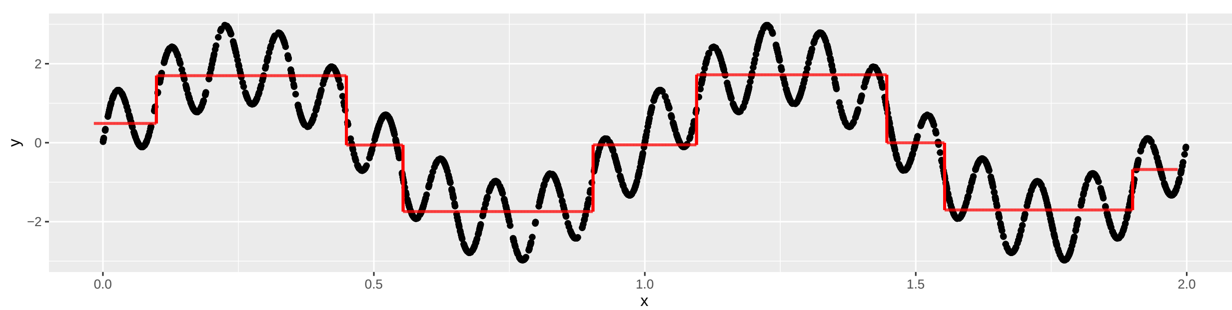
(c) [3 pt / 43 pts] We fit g, standard CART model on $\mathbb{D}$ with $N_0 = 1,000$. Draw g's predictions for all x atop $\mathbb{D}$ below:



(d) [3 pt / 46 pts] We fit g, standard CART model on a *bootstrap sample* of $\mathbb{D}$ with $N_0 = 1,000$. Draw g's predictions for all x atop $\mathbb{D}$ below:



(e) [4 pt / 50 pts] We fit g the best CART model on $\mathbb{D}$ with maximum *nine* leaves. Draw g's predictions for all x atop $\mathbb{D}$ below:



Problem 3 Assume that $\mathcal{Y} = \{0, 1\}$. Consider the following algorithms: bagged trees ($\mathcal{A}_{BAG}$), Random Forest ($\mathcal{A}_{RF}$) and gradient-boosted trees ($\mathcal{A}_{GBT}$). All three algorithms share $M = 1,000$ otherwise assume the other hyperparameters (which are specific to each of these algorithms) were optimized using model selection. Denote the resultant models generated from $\mathbb{D}$ as g_{BAG}, g_{RF}, g_{GBT}.

(a) [15 pt / 65 pts] Which one(s) of the following statement(s) are most likely true?

i) All three algorithms will perform equally out of sample as measured by misclassification error.

ii) TRUE The base / constituent models of each of these algorithms are trees.

iii) The base / constituent models of each of these algorithms are fit with the CART algorithm exactly.

iv) g_{BAG}, g_{RF}, g_{GBT} are all sum-of-trees models.

v) g_{BAG}, g_{RF}, g_{GBT} are all average-of-trees models.

vi) TRUE g_{BAG}, g_{RF} are both average-of-trees models.

vii) The first base / constituent model of g_{BAG}, g_{RF}, g_{GBT} are fit with the same observations.

viii) The first base / constituent model of g_{BAG}, g_{RF} are fit with the same observations.

ix) TRUE The base / constituent models of g_{BAG}, g_{RF} are likely fit with $N_0 \approx 1$.

x) The base / constituent models of g_{BAG}, g_{RF} are likely fit with N_0 large.

xi) The base / constituent models of g_{GBT} are likely fit with $N_0 \approx 1$.

xii) TRUE The base / constituent models of g_{GBT} are likely fit with N_0 large.

xiii) The base / constituent models of g_{BAG} have high bias, low variance

xiv) The base / constituent models of g_{RF} have high bias, low variance

xv) TRUE The base / constituent models of g_{GBT} have high bias, low variance

Problem 4 Consider the following data frame which has three metrics measured on five observations:

A	B	C
77516	13	2174
83311	13	0
215646	9	0
NA	7	NA
NA	NA	0

(a) [2 pt / 67 pts] If we were to use listwise deletion, how many rows would be left?

3

(b) [3 pt / 70 pts] If you were using this data for prediction in the future, would you modify it more than just imputing the values? If yes, explain how you would further modify it. If no, explain why not.

Yes. You would add three new columns that record 1 = missing, 0 = not missing for each of the three feature columns.

(c) [1 pt / 71 pts] Would the `missForest` algorithm be able to impute the values in the original dataset? Yes / no

(d) [4 pt / 75 pts] If we were to use imputation via averages, provide the fully imputed table below. Round to one decimal point.

A	B	C
77516	13	2174
83311	13	0
215646	9	0
$\boxed{125491}$	7	$\boxed{543.5}$
$\boxed{125491}$	$\boxed{10.5}$	0

(e) [1 pt / 76 pts] Your imputed data set is in which format? Circle one: wide or long

(f) [5 pt / 81 pts] If you circled "wide" in the previous question, then convert the table to long below. If you circled "long" in the previous question, then convert the table to wide below. Label the columns in your converted table appropriately.

metric	val
A	77516
A	83311
A	215646
A	125491
A	125491
B	13
B	13
B	9
B	7
B	10.5
C	2174
C	0
C	0
C	543.5
C	0

Note: the order of the rows does not matter.

Problem 5 Consider a dataset $\boldsymbol{X}$ of size $n \times p$ and $\boldsymbol{y} \in \mathbb{R}^n$ such that all y_i values are unique. Assume that there is at least one feature that can reduce ignorance error.

Consider fitting a regression tree model. Let the number of nodes in the tree be denoted κ. We will use the CART algorithm with one twist. Instead of running CART straight through which outputs the final, optimized tree, we instead proceed step-by-step and record the intermediate candidate trees to get a sequence of trees terminating the tree that CART returns.

To create this sequence, we begin with a single root note which predicts $\bar{y}$. Since there's one node, $\kappa = 1$. Then we make the best split based on weighted SSE across the left and right (the same objective function used in the CART algorithm). Because we now have a left and right leaf node, $\kappa = 3$. Now we try to split the left node and try to split the right node and use whichever split is better. There are now two more nodes in the tree so $\kappa = 5$. Every time a split occurs, κ increases by 2.

Thus, the candidate trees have values of $\kappa \in \{1, 3, 5, \ldots, M\}$ where M is the number of nodes when the CART algorithm terminates. Let the tree models be denoted $g_1, g_3, g_5, \ldots, g_M$.

Assume $N_0 = 1$ and $n > 100$.

(a) [9 pt / 90 pts] Which one(s) of the following statement(s) are most likely true?

i) TRUE M must be finite

ii) M will be the same value for all $\mathbb{D}$ with n examples (that is, you can always derive the value of M as a function of n only).

iii) M will be the same value for all $\mathbb{D}$ with p features (that is, you can always derive the value of M as a function of p only).

iv) TRUE The tree where $\kappa = 1$ is most likely underfit.

v) The tree where $\kappa = 1$ is most likely overfit.

vi) The tree where $\kappa = M$ is most likely underfit.

vii) TRUE The tree where $\kappa = M$ is most likely overfit.

viii) TRUE The best tree for oos prediction *could be* the tree where $\kappa = 5$

ix) The best tree for oos prediction *could be* the tree where $\kappa = M$

Now we want to select the best candidate tree from this sequence of trees. We will use the forward stepwise procedure and test all candidate trees in order of κ beginning with 1, then 3, etc. We will split $\mathbb{D}$ into $\boldsymbol{X}_{train}, \boldsymbol{y}_{train}$ and $\boldsymbol{X}_{select}, \boldsymbol{y}_{select}$ randomly. Let $\kappa^\star$ denote the number of nodes in the stewpise procedure's selected tree model.

(b) [12 pt / 102 pts] Which one(s) of the following statement(s) are most likely true?

i) TRUE The set of values in the vector $[\boldsymbol{y}_{train}^\top \; \boldsymbol{y}_{select}^\top]$ are the same as the set of values in $\boldsymbol{y}^\top$

ii) oos performance for g_5 can be estimated by computing $(\boldsymbol{y}_{train} - g_5(\boldsymbol{X}_{train}))^2$

iii) oos performance for g_5 can be estimated by computing $(\boldsymbol{y}_{train} - g_5(\boldsymbol{X}_{select}))^2$

iv) TRUE oos performance for g_5 can be estimated by computing $(\boldsymbol{y}_{select} - g_5(\boldsymbol{X}_{select}))^2$

v) $\kappa^\star = \underset{\kappa}{\arg\max} \left\{(\boldsymbol{y}_{select} - g_\kappa(\boldsymbol{X}_{select}))^2\right\}$

vi) TRUE $\kappa^\star = \underset{\kappa}{\arg\min} \left\{(\boldsymbol{y}_{select} - g_\kappa(\boldsymbol{X}_{select}))^2\right\}$

vii) $\kappa^\star = \underset{\kappa}{\arg\max} \left\{\left(y_{select_\kappa} - g_\kappa(\boldsymbol{x}_{select_{\kappa \cdot}})\right)^2\right\}$

viii) $\kappa^\star = \underset{\kappa}{\arg\min} \left\{\left(y_{select_\kappa} - g_\kappa(\boldsymbol{x}_{select_{\kappa \cdot}})\right)^2\right\}$

ix) TRUE The sequence $(\boldsymbol{y}_{train} - g_1(\boldsymbol{X}_{train}))^2, (\boldsymbol{y}_{train} - g_2(\boldsymbol{X}_{train}))^2, (\boldsymbol{y}_{train} - g_5(\boldsymbol{X}_{train}))^2, \ldots$ is monotonically decreasing in expectation for all valid κ values

x) The sequence $(\boldsymbol{y}_{train} - g_1(\boldsymbol{X}_{train}))^2, (\boldsymbol{y}_{train} - g_2(\boldsymbol{X}_{train}))^2, (\boldsymbol{y}_{train} - g_5(\boldsymbol{X}_{train}))^2, \ldots$ is monotonically increasing in expectation for all valid κ values

xi) The sequence $(\boldsymbol{y}_{select} - g_1(\boldsymbol{X}_{select}))^2, (\boldsymbol{y}_{select} - g_2(\boldsymbol{X}_{select}))^2, (\boldsymbol{y}_{select} - g_5(\boldsymbol{X}_{select}))^2, \ldots$ is monotonically decreasing in expectation for all valid κ values

xii) The sequence $(\boldsymbol{y}_{select} - g_1(\boldsymbol{X}_{select}))^2, (\boldsymbol{y}_{select} - g_2(\boldsymbol{X}_{select}))^2, (\boldsymbol{y}_{select} - g_5(\boldsymbol{X}_{select}))^2, \ldots$ is monotonically increasing in expectation for all valid κ values

(c) [5 pt / 107 pts] Do you need to modify this procedure in order to provide an oosSSE performance estimate for $g_{\kappa^\star}$? Yes / no and explain how it is modified and how this oosSSE performance estimate for $g_{\kappa^\star}$ is computed.

Yes. Even though you are assessing the best of the tree models on $\mathbb{D}_{select}$, you could be overoptimizing for $\mathbb{D}_{select}$. You need a separate hold-out set of observations called $\mathbb{D}_{test}$ so thus $\mathbb{D}$ should be split into $\boldsymbol{X}_{train}, \boldsymbol{y}_{train}$, $\boldsymbol{X}_{select}, \boldsymbol{y}_{select}$ and $\boldsymbol{X}_{test}, \boldsymbol{y}_{test}$ randomly. Then you can compute the oosSSE via $(\boldsymbol{y}_{test} - g_{\kappa^\star}(\boldsymbol{X}_{test}))^2$.

Problem 6 This problem is about the randomness in the bias-variance decomposition formula. Recall the four equations we studied in class:

$$\begin{aligned} Y_i &= f(\boldsymbol{X}_{i\cdot}) + \Delta_i, \ \forall i \\ G &= \mathcal{A}(<\boldsymbol{X}_{1\cdot}, \ldots, \boldsymbol{X}_{n\cdot}, Y_1, \ldots, Y_n>, \mathcal{H}), \\ \hat{Y}_\star &= G(\boldsymbol{X}_{\star\cdot}))^2, \\ MSE &= \mathbb{E}\left[(Y_\star - \hat{Y}_\star)^2\right] \end{aligned}$$

(a) [4 pt / 111 pts] Let the element of set $R = \{\Delta_\star\}$ be random but nothing else. Which quantity(ies) are random in the above four equations besides the element in the given set R? In the MSE formula, assume the expectation is taken over the set R. If the answer is the empty set, write "none".

$Y_\star, MSE$

(b) [3 pt / 114 pts] Let the elements of set $R = \{\Delta_\star, \boldsymbol{X}_{\star\cdot}\}$ be random but nothing else. Which quantity(ies) are random in the above four equations *in addition to* the answers from (a) besides the elements in the given set R? In the MSE formula, assume the expectation is taken over R. If the answer is the empty set, write "none".

$\hat{Y}_\star$

(c) [3 pt / 117 pts] Let the elements of set $R = \{\Delta_\star, \boldsymbol{X}_{\star\cdot}, \Delta_1, \ldots, \Delta_n\}$ be random but nothing else. Which quantity(ies) are random in the above four equations *in addition to* the answers from (b) besides the elements in the given set R? In the MSE formula, assume the expectation is taken over R. If the answer is the empty set, write "none".

$Y_1, \ldots, Y_n, G$

(d) [3 pt / 120 pts] Let the elements of set $R = \{\Delta_\star, \boldsymbol{X}_{\star\cdot}, \Delta_1, \ldots, \Delta_n, \boldsymbol{X}_{1\cdot}, \ldots, \boldsymbol{X}_{n\cdot}\}$ be random but nothing else. Which quantity(ies) are random in the above four equations *in addition to* the answers from (c) besides the elements in the given set R? In the MSE formula, assume the expectation is taken over R. If the answer is the empty set, write "none".

none